Research Design

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What is Research Design ?

- A research design is a basic plan that guides the type of information to be collected, the source of data and phase of analysis of the research project.
- The research design constitutes the blueprint for the collection, measurement, and analysis of data.
- We can also say overall planning before doing investigation.

Classification of Research Design

I. Survey method
II. Exploring study
III. Descriptive research design
IV. Experimental research design
V. Time series

Survey Method:

A survey is a method of gathering information from a number of individuals. This method is very much popular among researchers for collecting and analyzing facts, opinions, and interpretations of respondents, individuals or organizations that possess the relevant information.

Types of Survey:

- Online survey.
- Paper survey.
- Telephonic surveys.
- One-to-one interviews.

Exploratory Study:

- The exploratory research is concerned with the discovery of a general form of the problem. Such kind of research requires flexibility. Usually researchers works informally but use tentative guideline.
- This survey is used to know about new ideas and relationships.

Example:

- Tribal people in Bangladesh have some peculiar and uncommon background characteristics that distinguish them from the rest of the population.
- Random information suggests that they have a large family size, low age at marriage, and high mortality.
- The planned study is explorative, which will help the researchers to formulate objectives and hypotheses, keeping in view the above peculiarities regarding their distinct demographic characteristics.

Descriptive Research Design:

- A descriptive study is undertaken in order to ascertain and to be able to describe the characteristics of variables in a situation.
- The goal of a descriptive study, is to describe relevant aspects of the phenomena of interest to the researcher from an individual, organization, industry or other perspective.

Example:

- For example, in a supermarket, a researcher can from afar monitor and track the customers' selection and purchasing trends.

Experimental Research Design:

Experimental designs are set up to study possible cause → effect relationships among variables. These are in contrast to correlational studies, Which examine the relationships among variables without necessarily trying to establish which causes another.

Example:

- An example of an experimental research design might be to investigate the effects of a new medication on blood pressure. In this study, participants would be randomly assigned to two groups: a treatment group that receives the new medication, and a control group that receives a placebo. The independent variable would be the medication (or placebo), while the dependent variable would be the participants' blood pressure.

Time Series:

A time series is a set of data points collected over a period of time, typically at regular intervals. Time series data is often used to analyze trends, patterns, and relationships between variables over time.

For example:

- Stock prices.
- Weather data.
- Economic indicators.
- Sales data

Research Design

Three steps are to be noted in research design.

1. What is a research design
2. Basic issue of a research design
3. Components of a research

1. What is a research Design:

Research design is always necessary before the research is conducted.

- Good research design
- Objectivity
- Validity
- Reliability
- Generalization

2. Basic issue of a research Design:

There are three basic issues of a research design:

- Type of investigation
- Purpose of the study
- Study setting

Type of investigation:

Casual: When the researcher wants to describe the cause of the problem, then the study is called a casual study.

Non-causal (correlational study): When the researcher is interested in describing the important variables that are associated with the problem then it is known as non-causal or correlational study.

Purpose of the study:

Exploratory: The exploratory research is conducted with the discovery of a general form of the problem. Exploratory studies are necessary since we do not currently know if there are differences is common styles, interpretation scheme, superior-subordinate relation expectations, and the skills.

Descriptive: Descriptive study analysis the characteristics of availability in particular situation.

Purpose of the study:

Hypothesis Testing: Hypothesis testing means study in hypothesis, which explain certain relationship among the different variable. In other words, it is a procedure to test the statement about the population on the basis of sample information.

Study setting:

There are two types of study setting.

Non-contrived study setting: When the study is conducted in natural environments, where events normally occur, is non-contrived settings. Correlational studies are conducted in non-contrived settings.

Study setting:

<u>Contrive study setting:</u> In research, a contrived study setting is a research environment that has been intentionally created or manipulated by the researcher to observe a specific phenomenon or behavior under controlled conditions.

Contrived study settings are often used in laboratory-based research where the researcher has control over the variables that may affect the outcome of the study.

Components of a research design

Any investigator, before getting on to the job of planning a research design for a proposed study, should be clear about the following aspects:

- What to observe
- Whom to observe
- How to observe
- Why to observe
- How to record the observation
- How to analyze the observation
- What inferences can be drwan

Reference Book:

- Ahmad,Nawaz."Research Methods. MA (Economics)": 55-64

Any Question ?

THANKS

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